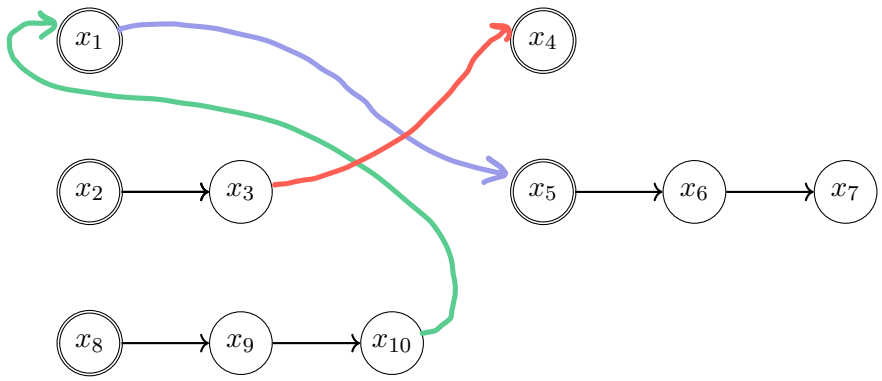
CSE2331

Union-Find Homework

For this entire assignment, you are not required to redraw the lists or trees.

Problem 1. The following linked lists represent disjoint sets,



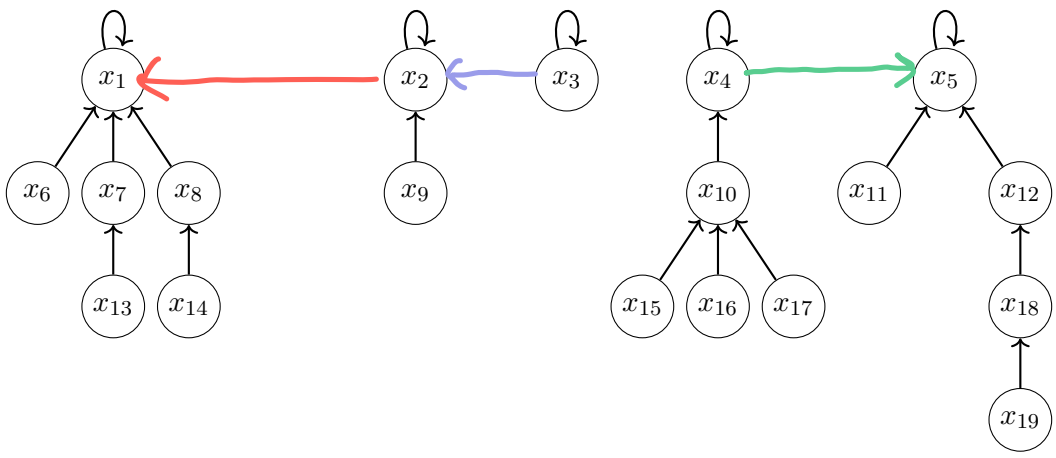
Consider the following sequence of Union-Find operations,

1: `WeightedUnion`(x_1,x_9) —
2: `FindSet`(x_1) x_8
3: `FindSet`(x_9) x_8
4: `WeightedUnion`(x_8,x_6) —
5: `FindSet`(x_1) x_8
6: `FindSet`(x_6) x_8
7: `WeightedUnion`(x_3,x_4) —
8: `FindSet`(x_1) x_8
9: `FindSet`(x_3) x_2

(a) What element is returned by each of the `FindSet` operations listed above?

(b) Draw the linked lists representing the final data structure.

Problem 2. The following trees represent five sets,



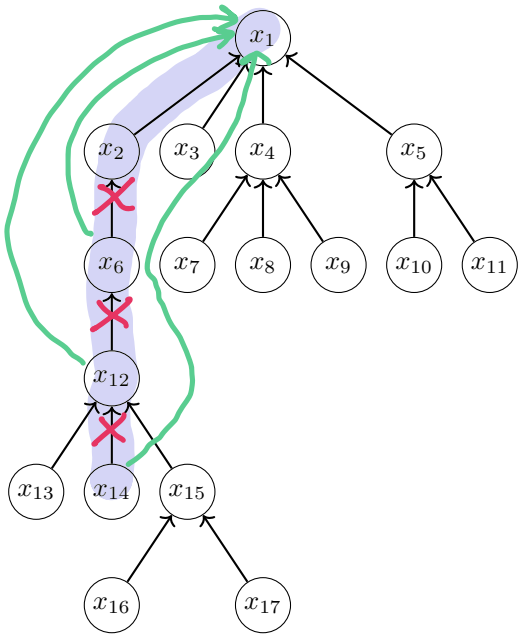
Consider the following sequence of Union-Find operations,

1: `UnionByHeight`(x_{16},x_{19})
2: `FindSet`(x_5) x_5
3: `FindSet`(x_{10}) x_5
4: `UnionByHeight`(x_2,x_3)
5: `FindSet`(x_3) x_2
6: `UnionByHeight`(x_3,x_{14})
7: `FindSet`(x_3) x_1
8: `FindSet`(x_6) x_1

(a) What element is returned by each of the `FindSet` operations listed above?

(b) Draw the trees representing the final data structure.

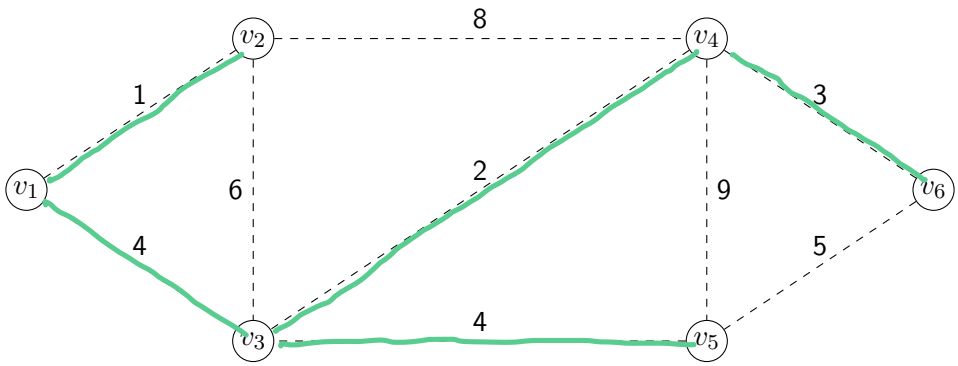
Problem 3. Consider the following tree representing a set in the Union-Find data structure,

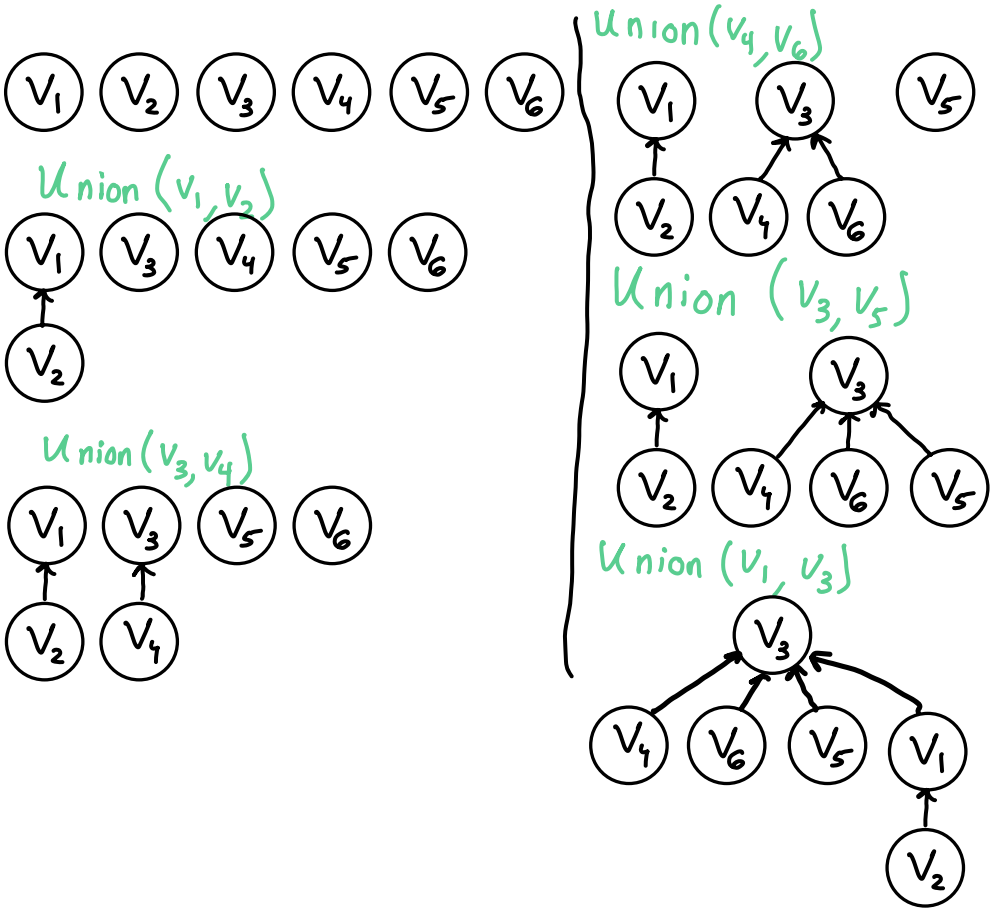


Run the `FindSet` operation on x_{14} with **path compression** and draw the resulting tree.

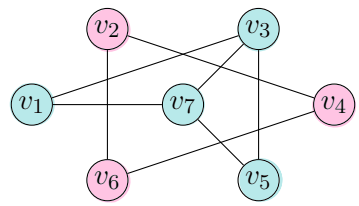
Problem 4. Revisit Kruskal's Algorithm from the Minimum Spanning Tree lectures. Find the minimum spanning tree of the following weighted graph using the algorithm. Be sure to:

- Whenever an edge is added to the spanning tree, denote what the current independent sets are and what union operation was performed.
- Whenever an edge cannot be added to the spanning tree because it would create a cycle, denote that the edge is being "considered, but ignored".
- Assume that the union find data structure that is used is implemented using a tree based structure. Draw the trees that represent the disjoint sets at every step of the algorithm.





Problem 5. As discussed in the notes, one application of Union-Find is to find the connected components of a graph. Consider the graph below



What are the connected components of the graph? Express the connected components of the graph as SETS. Note: This question is easy but this is an application of union find!

$\{v_1, v_3, v_5, v_7\}$

$\{v_2, v_4, v_6\}$